Computed tomography (CT) scans rarely require the presence of an anesthesia provider for an adult patient, but young children often require some form of anxiolysis. Studies have demonstrated that CT scans performed on children using general anesthesia resulted in atelectasis severe enough to obscure evidence of metastasis on the scan. Adding 5 cm H2O of PEEP was sufficient enough to prevent this from occurring.



Although the degree of ionizing radiation makes access to the patient limited, the patient is much more visible than a patient in an MRI suite. And, unlike an MRI, no special anesthesia equipment is required. General anesthesia can be performed with an LMA if the patient has no contraindications. If the patient has received oral contrast, however, an endotracheal tube must be used to prevent aspiration.

Propofol, midazolam, ketamine, and pentobarbital have been used successfully for conscious sedation for procedures such as CT scan and MRI.

Pregnancy is a contraindication to the administration of intravenous contrast media (ICM). Other significant, but less important factors in determining whether a patient can receive ICM include: advanced age, cardiac disease, multiple medical disorders, preexisting azotemia, and patients being treated with potentially nephrotoxic drugs such as nonsteroidal antiinflammatory agents and aminoglycoside antimicrobials. It is also important to inquire about any history of asthma, allergies, or prior allergic reactions to ICM.

The extravasation of ICM can result in tissue necrosis and sloughing of the skin, mandating keen observation of the IV site. The administration of as little as 1 mL of ICM can result in anaphylactic or anaphylactoid reactions resulting in hypotension, bronchospasm, and shock. Emergency measures for the treatment of anaphylactic-type reactions should be readily available whenever ICM is administered.

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